M.AI.ESTRO VR: system audit

Finalized 4 October 2026 · Graham Pellegrini · University of Malta

System Demo

Full recording stitch

Open full recording

Clock orbit: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right · Effective 3× speed (video encoded at 1.5×, player at 2×). Open orbit comparison

Clock task: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right. Task replay uses the latest orbit mesh under the temporary demo policy. Open task comparison

Pipeline Diagram

The pipeline PNG shows the flow from headset capture to stored object evidence, with separate task and orbit branches. Open the full-resolution image to inspect individual labels; the PDF gives the complete diagram its own landscape A3 page.

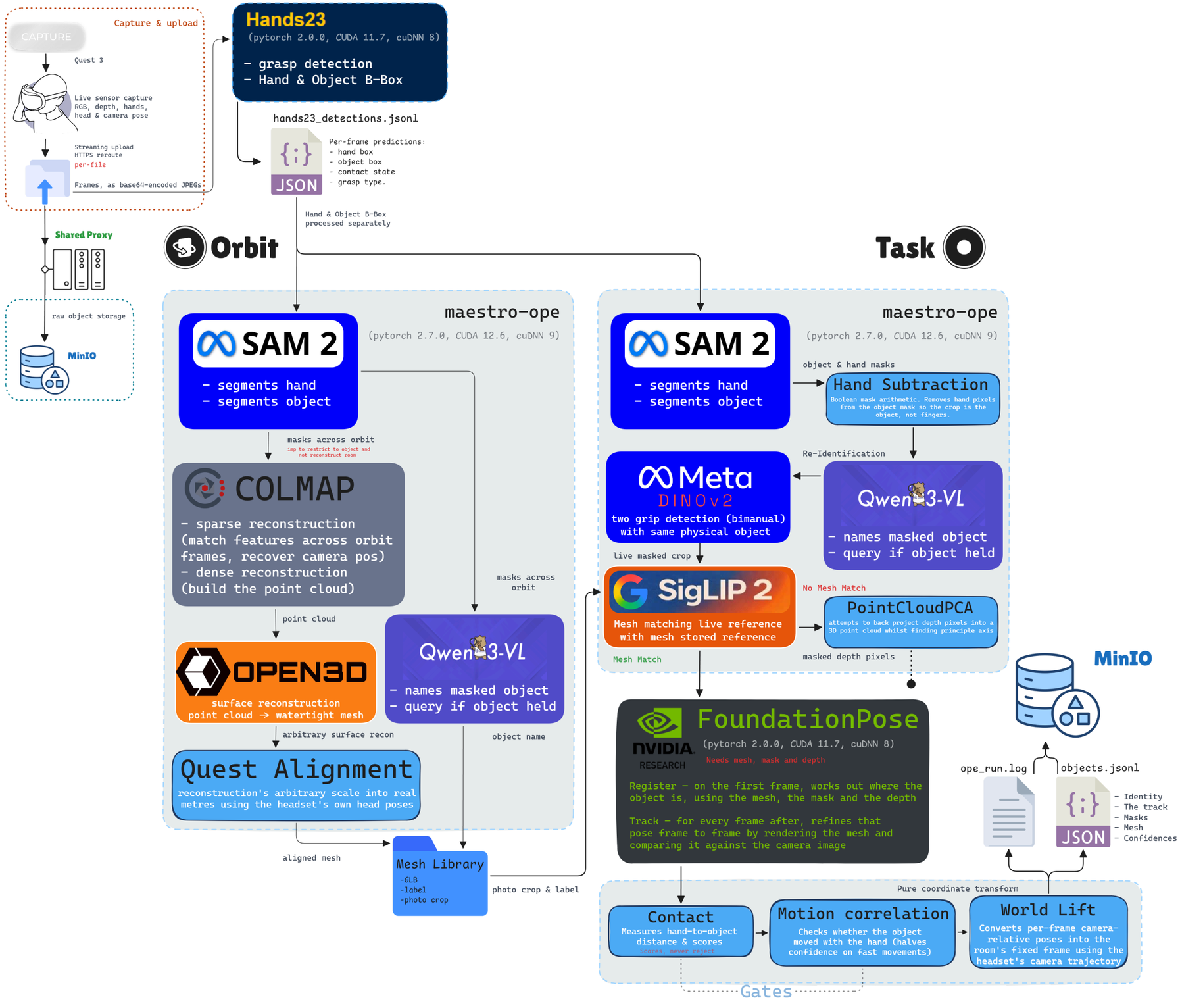


Figure 1. Current Maestro VR pipeline used for the audit discussion.

Pipeline Modules

Each node has a narrow responsibility. The audit question is whether the handoff between nodes is reliable enough for replay.

1

Quest capture

Records RGB, depth, headset pose and hand pose for task and orbit sessions.

Quest sensor capture

2

Ingestion and MinIO

Accepts uploads, stores raw inputs and exposes session artifacts for worker processing.

MinIO object storage

3

Hands23 hand detection

Provides hand/object contact evidence and candidate boxes before object processing begins.

Hands23 code and explanation

4

SAM2 segmentation

Turns candidate evidence into masks for orbit reconstruction and task-frame object tracking.

SAM2 overview and code

5a

PointCloudPCA

Provides model-free task pose estimates when a trusted mesh is not available.

Project pose measurements

5b

FoundationPose with mesh

Runs when a suitable mesh, mask and depth are available; this is the stronger route for mesh-backed replay.

FoundationPose overview and code

6

Qwen labels

Supplies an object label from the crop; useful, but still dependent on mask quality.

Qwen3-VL-4B-Instruct model

7

DINOv2 and SigLIP

Support visual identity checks. The current audit keeps DINOv2 and SigLIP rather than adopting DINOv3.

DINOv2 ViT-S/14 model · SigLIP base model

8

COLMAP mesh reconstruction

Reconstructs object meshes from orbit capture frames for canonical mesh-backed replay.

COLMAP tutorial and code

9

Headset replay

Consumes stored objects, tracks, meshes and masks so captured objects can be inspected in VR.

Unity XR Interaction Toolkit

System Status

Open items are always visible. Fixes made before the 5 Oct audit are folded away. Open up to two cards at once.

▶ **Pre-audit fixes (before 5 October)**

Orbit capture and 3D model generation

An orbit must follow one object through turns and flips, then build a complete 3D model, underside included.

Modules 3, 4, 8, 9 **In progress**

What works

- A deliberate first grip starts the orbit (hand selection is under Pre-audit fixes).
- Two 3D models, a candle and a clock, were built from orbits by hand in September, with real size from the headset's own movement.

Problems found

- **Flipping the object mid-orbit.** The tracker loses the object or makes duplicates. The bottle kept its object but added 4 duplicate candidates. In the candle and stopwatch orbits no object was found before the flip.
- **The clock underside.** Both trackers lose the clock when its underside faces the camera. Likely cause: the orbit and the model never covered the underside. Not yet proven.
- **No 3D model is built after an orbit.** Nothing starts it. 11 orbits from Batch 2B are waiting. The first test on a new orbit (wooden clock) failed because the mask covered only the dial.
- **Frames without a mask** make the model builder stop instead of skipping them.
- **Headset replay.** Facing the viewer at spawn is still open. The slanted start of the later clock model comes from the server-built model. Placement and size of new recordings, and grabbing with a distance pointer, are only partly done. FoundationPose replay on the headset is unverified.

Next

- Lock onto one object from the first grip, merge duplicates, and treat later turns and flips as the same object.
- A model builder that copes with a moving object, with a single-photo generator as a fallback marked "needs review".
- Build the model automatically after each orbit. Every model is checked by a person before it joins the library.

Still to investigate

- Whether flipped orbits can be rebuilt at all, or must be split into rigid parts.
- Why the underside is lost: the model, the tracker, or both.

Mesh matching and selection

A first matcher is on for three objects. New objects have no model, and replay still uses a temporary shortcut.

Modules 7, 9 **In progress**

Where it stands

- Each clip is compared with reference recordings. Settings are fixed and were not tuned. It is on for the ceramic clock, the scented candle and the rolling pin.
- More reference recordings cut wrong matches from 8 to 0 in a replay of 107 objects (counted by reading, not confirmed labels). That larger library is not live.
- None of the 43 Batch 2B clips matched, because those objects have no model.
- Replay shortcut: every task object is served the latest orbit model without an identity check.

Next

- Add approved models to the library with a record of where each came from.
- Match again against new models once they exist.

Still to investigate

- Look-alike objects that match by mistake. Real accuracy needs fresh recordings, not the clips used to design it.

No-mesh matching issue

Proposed: stop using the old point-cloud tracker as the automatic fallback for objects without a model. Use a manual trigger when needed.

Module 5a **Waiting confirmation**

The issue

- Without a model, the old point-cloud tracker estimates the object from the visible depth points. When the visible patch changes the estimate jumps. Largest jumps measured: 0.15 m (slow candle), 1.05 m (fast clock), 0.74 m (cup on table).
- Steadier-pose versions (W2, W3) cut the shake but locked in wrong directions. Not adopted.

Proposed decision

- Drop this route as an automatic path. Run FoundationPose only on objects that have a model, and trigger the no-model case by hand when it is needed.
- **Waiting for confirmation** that supervisors and colleagues agree with this direction.

Still to investigate

- Whether W2 or W3 add anything on top of FoundationPose. Not tested.

FoundationPose tracking

Beat the old tracker on four clock clips. In the live worker it runs beside the old tracker and does not replace it yet.

Module 5b **In progress**

Where it stands

- Our earlier description overstated how far it was used in the live pipeline. It now runs as a memory-capped service.
- In the live worker it runs in shadow mode: the old point-cloud pose is still the one saved. It skips orbit recordings.

- Same clips, same model, side by side: better on all four. No smoothing on either side. Both lose the clock when its underside faces the camera.

Next

- Decide when it becomes the main pose instead of a shadow, once the no-model decision is confirmed.

Still to investigate

- A confirmed replay on the headset. The earlier test showed only a wireframe.

FoundationPose overview and code

Object names

Names come from a fixed keyword list. Anything outside it shows as "unidentified".

Modules 2, 6 **In progress**

Where it stands

- Rubik's cubes, the book, the controller, tape and the cereal box all show as unidentified. Qwen also mislabels some objects (the candle as "hand watch").
- Labels also feed the rule that merges an object seen by both hands, so a wrong label can split it.

Next

- Name objects from Qwen's own description of what is being handled, with safe names and a fallback.

Empty and split results

10 clips came out empty and several objects were split in two.

Modules 3, 4 **In progress**

Where it stands

- Empty clips: 2 seed aborts, 4 quality-gate failures, 4 with no accepted grip. All 10 have hand-detector boxes on many frames.
- Splits: weak visual match, a same-hand-only merge rule and grip gaps just over half a second.
- Mask cleaning: using the current frame's hand box gave a small gain (switched on as a recorded exception). Restarting the mask did nothing.

Next

- Start the mask from the object box alone for clips that never started.
- One rule set that keeps the same object as one track, which also fixes the orbit duplicates.

Still to investigate

- How to start tracking small objects that are never found.
- Whether the size of the object changes how well the detector boxes it.

Shared GPU and the live worker

Other jobs filled the GPU our worker used. Maestro now has its own GPU, and a lock keeps heavy jobs apart.

Modules 2, 5b, 8 **In progress**

Where it stands

- On 12 Oct other jobs filled GPU 2, so any new upload would have failed. All Maestro services now run on GPU 3, tested end to end.
- The old pose server is stopped. Services are named and labelled so anyone can see who holds the GPU.

Next

- A lock so clip processing, pose registration and 3D model building take turns. The first test failed on a setting and on a process-detection step. It is being changed so a lock fault can never fail a session, then retested.

Still to investigate

- The real peak memory of a processing session, which is still a guess.

Dedicated Maestro server endpoint

Awaiting IT approval. The request is submitted and there has been no response yet.

Module 2 **Awaiting IT**

Where it stands

- Maestro uses the shared university proxy. A dedicated endpoint is wanted for EEAI 2026 and the edge-AI presentation.
- After IT supplies it, headset URLs, HTTPS, uploads, metadata and mesh downloads must be tested together.

~~Estimated Timeline~~

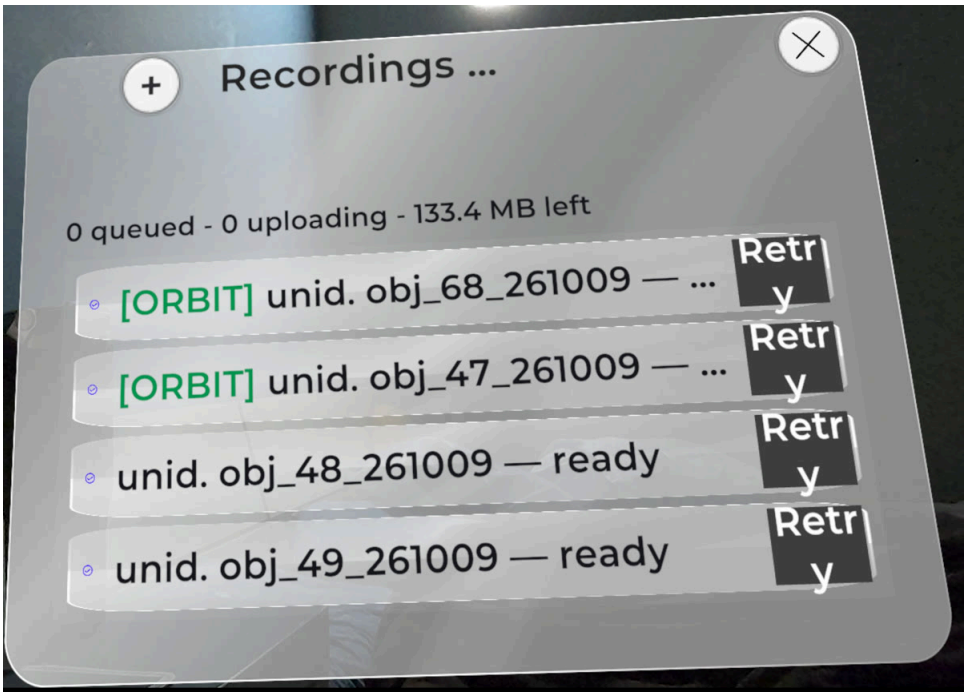
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Audit (12 October 2026)

Recordings

- Batch 2 was split into 2A and 2B. Uploads kept failing, so we built an upload queue first (first in, first out, with visible progress)
- 2B: 43 clips processed. None matched a 3D model, because no 3D models exist yet for the new objects. 10 came out empty and 4 split one object into two

Headset Recordings tab. "133.4 MB left" is one old failed clip being counted. A display fix is written and not yet installed



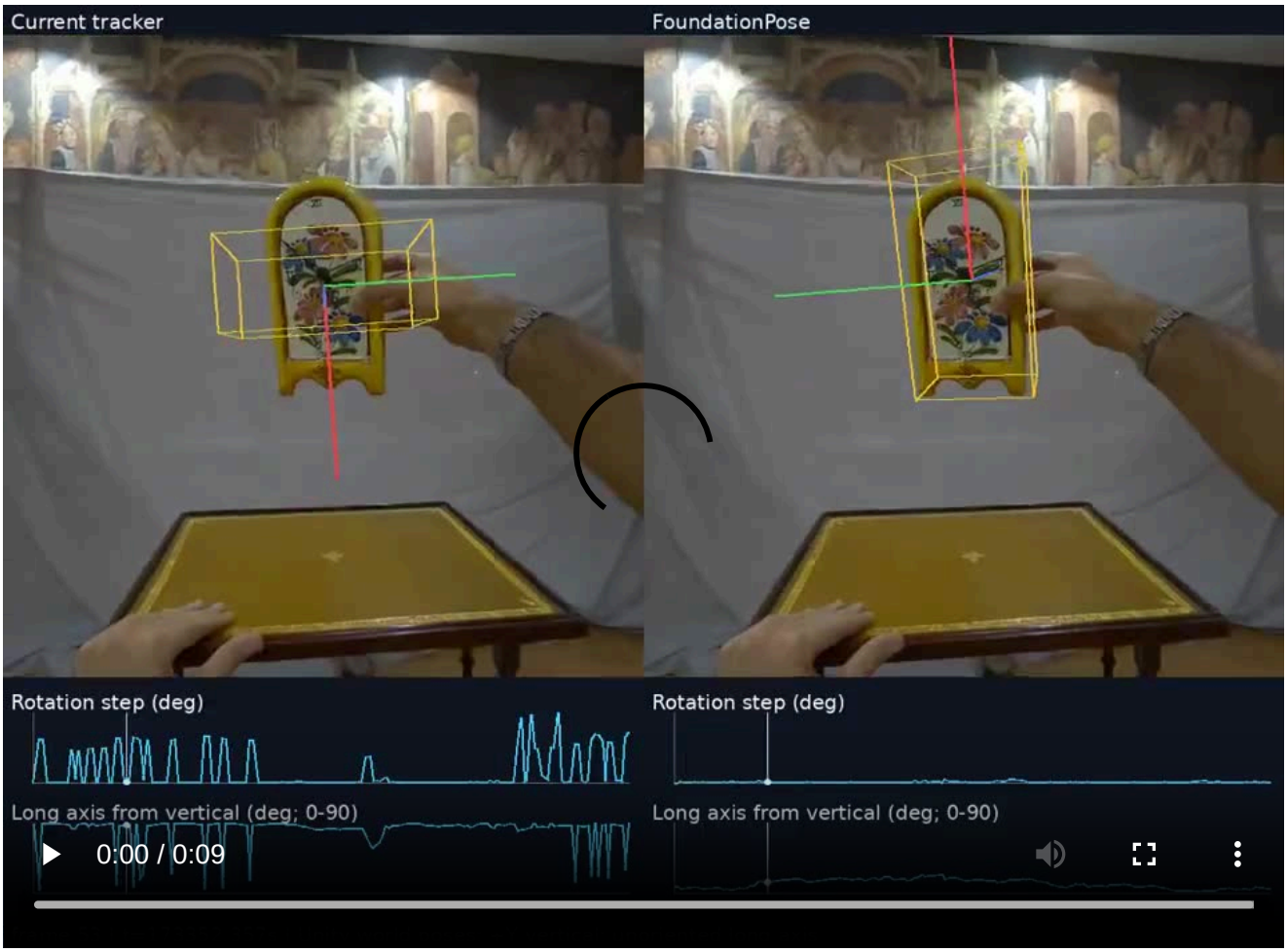
Old tracker vs FoundationPose

- FoundationPose tracks using the object's 3D model. The old tracker does not
- **No smoothing on either side**

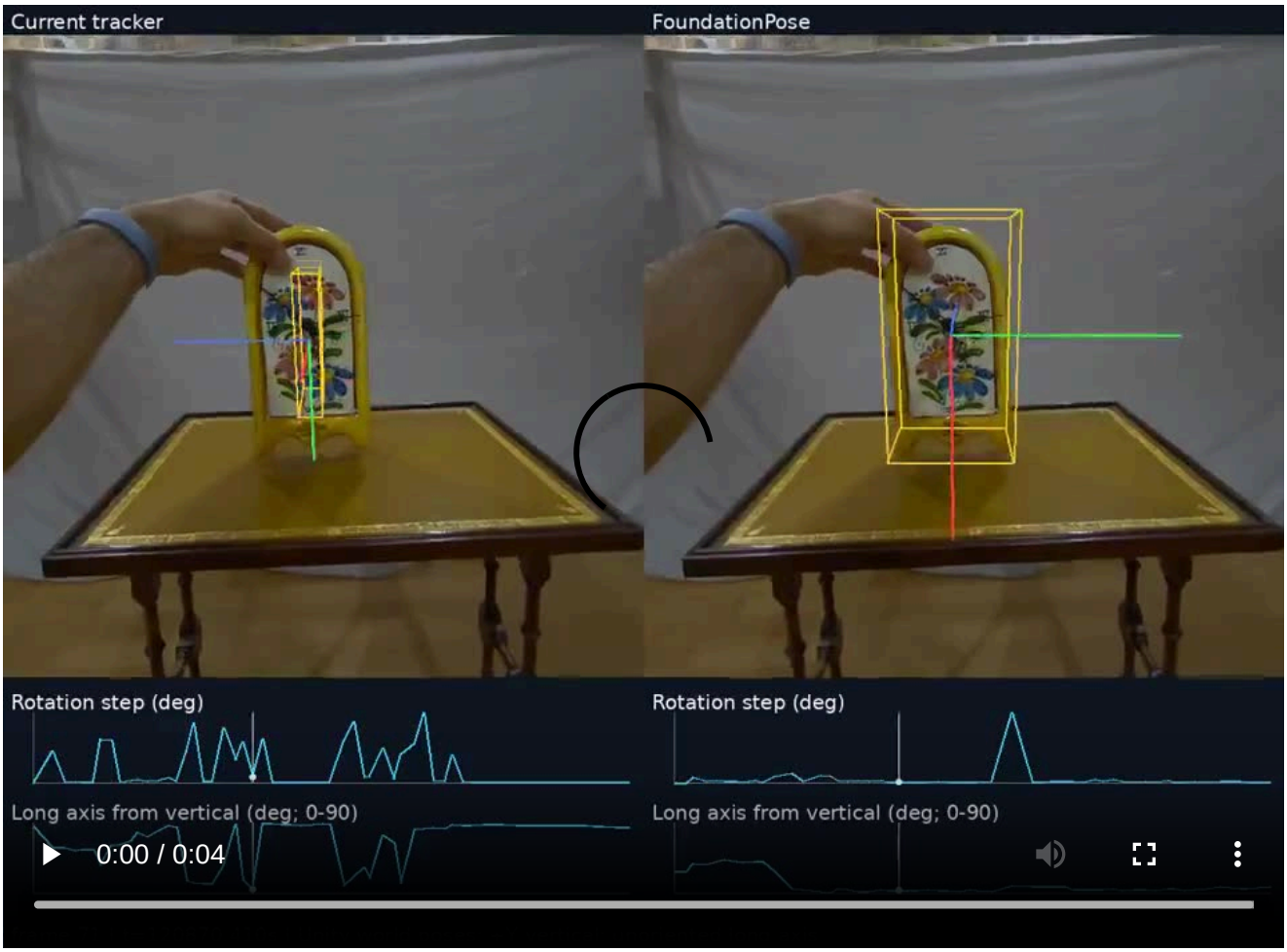
Clock, live clip. The old tracker spikes. FoundationPose stays upright



Clock, hand task



Clock, back view



Clock plus a second object. Shows the one-object-becomes-two problem



- **Problem: both trackers lose the clock when its underside faces the camera.** Likely an orbit issue, which we are investigating

Flipping the object mid-orbit

- The tracker loses the object and makes duplicates
- Flipping sometimes does not work at all: the candle and the stopwatch were not found
- The bottle kept the object but added 4 duplicates

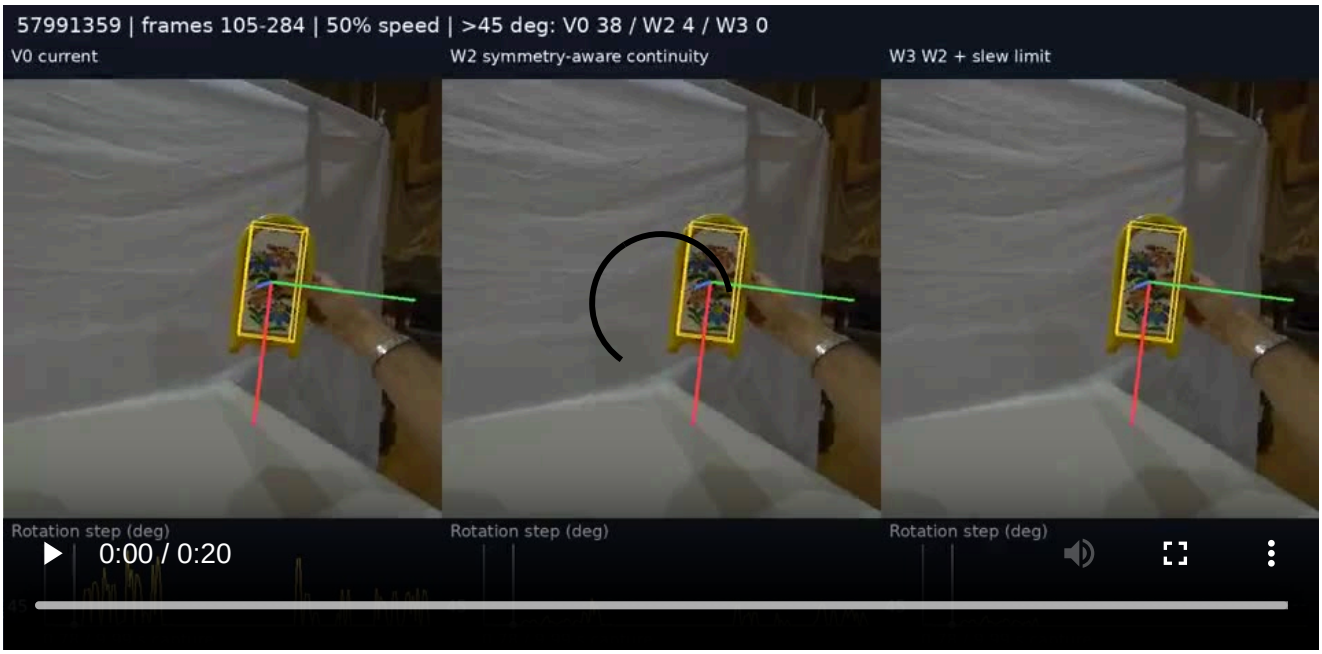
Bottle, flipped



Steadier pose (W2, W3)

- **W2** = symmetry-aware continuity (keeps the box from jumping between look-alike orientations)
- **W3** = W2 plus a speed limit on turning
- Done offline, alongside the FoundationPose work, on the old tracker's pose
- Result: much less shake, but it can lock in the wrong direction. On one clip it ended turned 90 degrees
- Not adopted for the old tracker
- **Open:** whether W2 or W3 help on top of FoundationPose. Not tested yet. FoundationPose's 3D model already fixes direction, so the gain may be small

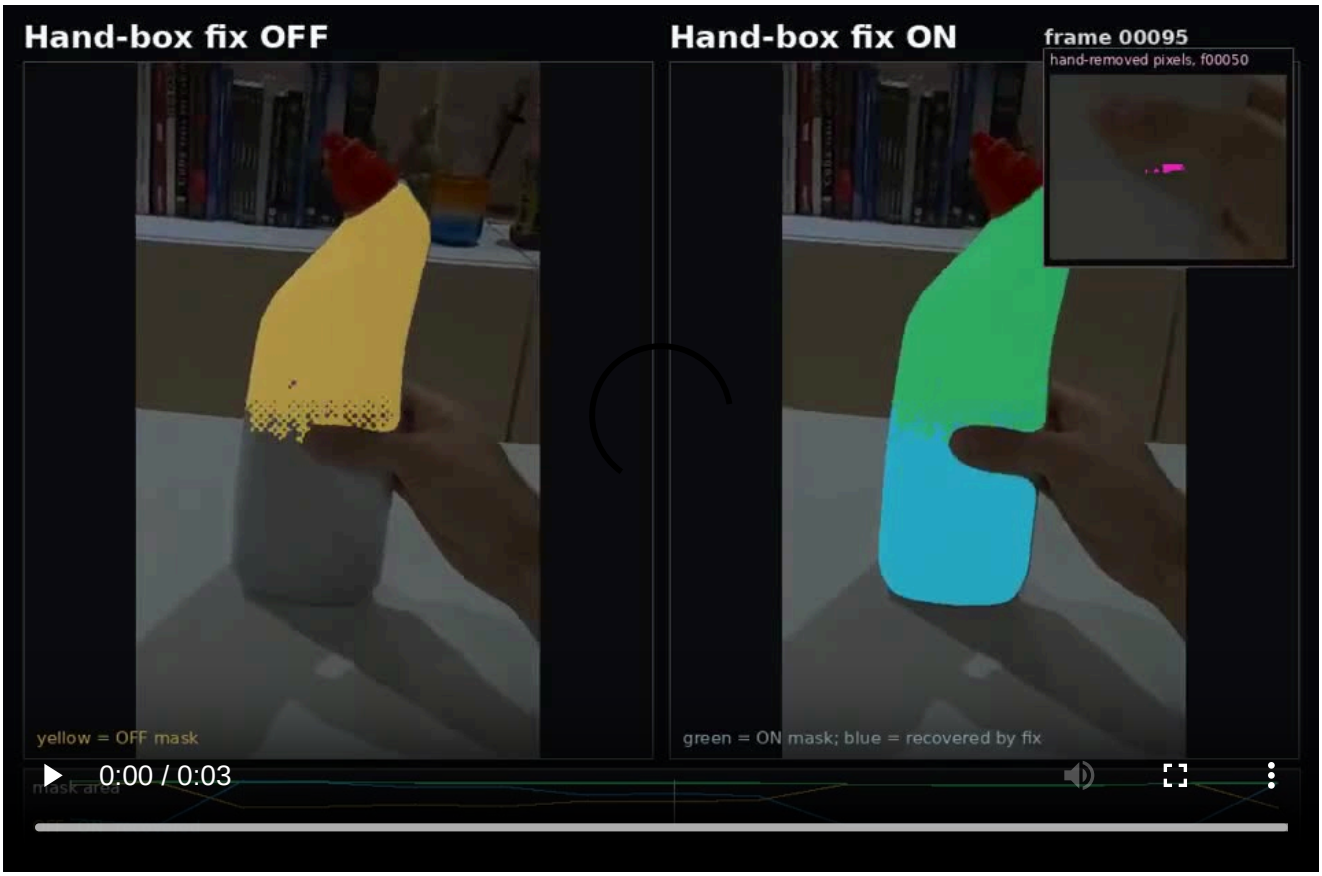
Current, W2, W3 on the same clip. Shake gone, W2 ends wrong



Cleaning the masks

- The mask is the outline of the object, with the hand cut out of it
- **Fix 1: cut the hand out using this frame's hand box.** Small gain. Fewer broken outlines. Not enough to adopt
- **Fix 2: restart the mask when it breaks.** No effect
- Still a problem: many clips never find the object or start badly. That needs a better start, not cleaning

Fix 1 off (left) and on (right), best case. The old version loses the lower half of the bottle (blue = what the fix recovers). The typical gain is much smaller



Matching

- Each clip is compared with reference recordings. The settings were not tuned
- More reference recordings helped. Wrong matches in a replay of 107 objects: 8, then 2, then 0 (counted by reading, not confirmed labels)
- Matching only helps once an object has a 3D model. That is why none of the 2B clips matched

Technical Detail

Full technical audit and measurements

Graham Pellegrini | M.AI.ESTRO VR | System audit | Finalized 4 October 2026